

The Δ -Method and the mgf Technique

- **Theorem 4.3.9.:** Let $\{X_n\}$ be a sequence of random variables such that

$$\sqrt{n}(X_n - \theta) \xrightarrow{D} N(0, \sigma^2).$$

Suppose the function $g(x)$ is differentiable at θ and $g'(\theta) \neq 0$. Then

$$\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{D} N(0, \sigma^2 g'(\theta)^2).$$

- Brief Proof:

$$\sqrt{n}(g(X_n) - g(\theta)) \approx \sqrt{n}g'(\theta)(X_n - \theta)$$

- **Theorem 4.3.10.:** Let $\{X_n\}$ be a sequence of random variables with mgf $M_{X_n}(t)$ that exists for $-h < t < h$ for all n . Let X be a random variable with mgf $M(t)$, which exists for $|t| < h_1 < h$. If $\lim_{n \rightarrow \infty} M_{X_n}(t) = M(t)$ for $|t| < h_1$. Then $X_n \xrightarrow{D} X$.

If $\psi(n) \rightarrow 0$ as $n \rightarrow \infty$,

$$\left(1 + \frac{b}{n} + \frac{\psi(n)}{n}\right)^{cn} \longrightarrow e^{bc}, \quad \text{as } n \rightarrow \infty$$

$$\left(1 + \frac{1}{n}\right)^n \rightarrow e$$

$$\left(1 + \frac{b}{n}\right)^{cn} \rightarrow e^{bc}$$

$$\left(1 - \frac{t^2}{n} + \frac{t^3}{n^{3/2}}\right)^{-n/2} \rightarrow e^{t^2/2}$$

- Let Y_n have a distribution that is $b(n, p_n)$. Suppose that the mean $\mu = np_n$ is the same for all n .

$$\begin{aligned}M(t; n) &= \mathbb{E}(e^{tY_n}) = [(1 - p_n) + p_n e^t]^n = \left[1 + \frac{\mu(e^t - 1)}{n}\right]^n \\&\rightarrow e^{\mu(e^t - 1)},\end{aligned}$$

which is Poisson distribution with mean μ .

- Whenever a random variable has a limiting distribution, we may, if we wish, use the limiting distribution as an approximation to the exact distribution function. We may use Poisson distribution as an approximation to the binomial distribution when n is large p is small.

- Poisson Approximation: Y has a binomial distribution with $n = 50$ and $p = 1/25$. Then

$$P(Y \leq 1) = \left(\frac{24}{25}\right)^{50} + 50 \frac{1}{25} \left(\frac{24}{25}\right)^{49} = 0.400.$$

Since $\mu = np = 2$, the Poisson approximation is

$$e^{-2} + 2e^{-2} = 0.406.$$

- Let Z_n be $\chi^2(n)$. Then the mgf of Z_n is $(1 - 2t)^{-n/2}$, $t < 1/2$. The mean and variance of Z_n are, respectively, n and $2n$. Find the limiting distribution of $Y_n = (Z_n - n)/\sqrt{2n}$.

$$\begin{aligned}M(t; n) &= \mathbb{E}\left\{\exp\left[t\left(\frac{Z_n - n}{\sqrt{2n}}\right)\right]\right\} = e^{-tn/\sqrt{2n}}\mathbb{E}(e^{tZ_n/\sqrt{2n}}) \\&= \exp\left[-\left(t\sqrt{\frac{2}{n}}\right)\left(\frac{n}{2}\right)\right]\left(1 - \frac{2t}{\sqrt{2n}}\right)^{-n/2} = (e^{t\sqrt{2/n}} - t\sqrt{\frac{2}{n}}e^{t\sqrt{2/n}})^{-n/2} \\&\rightarrow e^{t^2/2}.\end{aligned}$$

So, $Y_n = (Z_n - n)/\sqrt{2n}$ has a limiting standard normal distribution $N(0, 1)$.

- In the last example, we have

$$\sqrt{n}\left[\frac{1}{\sqrt{2n}}Z_n - \frac{1}{\sqrt{2}}\right] \rightarrow^D N(0, 1).$$

Let $g(t) = \sqrt{t}$ and Let $W_n = g(Z_n/\sqrt{2n}) = (Z_n/\sqrt{2n})^{1/2}$. Note that $g(1/2) = 1/2^{1/4}$ and $g'(1/\sqrt{2}) = 2^{-3/4}$. By the Δ -method, we have

$$\sqrt{n}(W_n - (1/2)^{1/4}) \xrightarrow{D} N(0, 2^{-3/2}).$$